

PREET PATEL

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TECHNICAL GENERALIST & SOLUTIONS PROFILE

Technical generalist with 4+ years of experience turning ambiguous, high-consequence information into structured requirements, evidence-backed findings, and practical technical work. Background spans AI product evaluation, physics and photonics, scientific software, statistical analysis, and stakeholder-facing technical review. Learns unfamiliar systems quickly and carries problems from definition through validation and communication.

TECHNICAL SOLUTIONS & PRODUCT EXPERIENCE

Mercor (Contractor)

Apr 2026 - Present

AI Evaluations - Generalist

- **LLM Evaluation:** Conduct human-in-the-loop evaluations of LLM-based products for reasoning quality, factual consistency, usability, and instruction adherence; isolate failure modes and document reproducible evidence using structured rubrics.

U.S. Patent & Trademark Office

Oct 2024 - Present

Patent Examiner (Physics - Photonics & Optical Systems)

- **Evidence-Based Review:** Synthesize claims, drawings, specifications, and technical literature into traceable written findings and defensible determinations under rigorous quality, compliance, and timeliness standards.
- **Stakeholder Communication:** Conduct structured interviews with applicants and counsel to clarify technical intent, explain findings, and identify practical paths to resolution while independently managing a high-volume docket.
- **Rapid Domain Learning:** Develop working knowledge of unfamiliar optical, quantum, and electronic systems quickly enough to evaluate detailed technical arguments and communicate precise conclusions.

GlobalLogic Inc. (Google RMI Team)

Jul 2024 - Oct 2024

Data Quality Analyst ("Superrater")

- **Testing & Feedback:** Designed complex prompts to probe model boundaries and translated observed failure patterns into actionable feedback for iterative product improvement.

SCIENTIFIC COMPUTING EXPERIENCE

University of California, Davis

Mar 2021 - Sep 2023

Computational Astrophysicist

- **Large-Scale Computing:** Built reproducible Python and HDF5 analysis pipelines for large FIRE-2 simulation datasets on national supercomputing systems, improving runtime by 50%+ through modular software and parallel workflows.
- **Technical Translation:** Converted open-ended research questions into tractable analytical plans, validation criteria, and decision-relevant visualizations for domain specialists.

SELECTED TECHNICAL WORK

Scientific Analysis Framework & Element Tracers

Python / HDF5 / HPC · GitHub

- Translated a fragmented scientific workflow into a reusable framework, balancing computational performance, documentation, and adoption by other researchers.

Hospitalist Burnout Study

Biostatistics / Survey Data · DOI

- Designed and reported nonparametric comparisons for a 43-participant hospitalist survey, translating ordinal response data into reproducible findings for a peer-reviewed publication.

EDUCATION

M.S. Physics - University of California, Davis

2020 - 2023

B.S. Physics & B.S. Astrophysics, Minor in Statistics - University of Michigan, Ann Arbor

2015 - 2019

CORE EXPERTISE

Solutions & Delivery: technical discovery, requirements clarification, workflow mapping, issue reproduction, root-cause analysis, technical documentation, stakeholder interviews, evidence-backed recommendations.

Software & Data: Python, NumPy, Pandas, SciPy, Matplotlib, SQL, C++, Git, Bash, Linux/Unix, HDF5, data visualization.